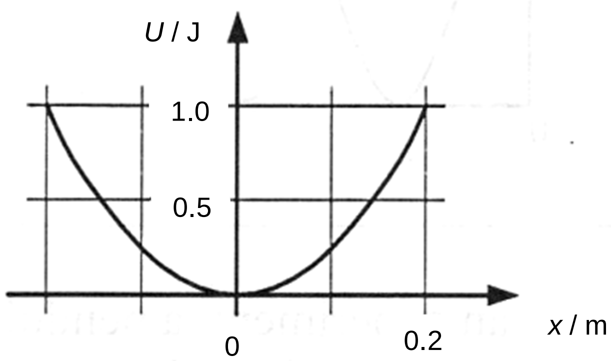
16 The given graph shows the variation with displacement *x* of the potential energy *U* of a particle of mass 4 kg moving in simple harmonic motion.



Which of the following is the period of oscillation of the mass?

A 0.3 s **B** 0.9 s **C** 1.1 s **D** 1.8 s

$$TE = KE_{\max} = \frac{1}{2}mv_o^2 = \frac{1}{2}m(\omega x_o)^2 = \frac{1}{2}m\frac{4\pi^2}{T^2}x_o^2$$

$$1 = \frac{1}{2}(4)\frac{4\pi^2}{T^2}(0.2)^2$$

$$T = \frac{2\pi\sqrt{2}}{5}\,s$$

Ans: D

17 The string shows the shape at a particular instant of part of a progressive wave travelling along